

ZX Diagrams at the Seam: Spiders, Pauli Webs, Gadgets, and the Cafeteria Problem of Cross-Disciplinary Imports

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Abstract

Diagrammatic languages are the most successful interface between quantum computing and the human mind. The ZX calculus — with its spiders, Pauli webs, and gadgets — is taught as *the* intuitive picture of quantum processes, and its completeness theorems are among the finest results in the field. Yet the same diagrams are increasingly being loaded with imports from disciplines that have nothing in common with a drawing plane: loop-quantum-gravity spin networks, quantum groups, holographic entropy, and toy black holes. This paper performs a fault-line audit of the three central constructs. It tables, for each construct, the exact import provenance — which discipline it came from, what it silently carries — and audits the mutual compatibility of those imports. The finding is precise: within the computational domain (circuits, codes, compilation) the imports are demonstrably coherent, and the diagrams earn every bit of their success; the fault lines appear exactly where 3+1-dimensional particle physics and 1-dimensional thermodynamics are mixed into a 2-dimensional map without any compatibility check. The paper then names the seam — the place where each diagram stops being a map and starts being mistaken for territory — and generalizes the resulting “cafeteria problem”: siloed disciplines remain separate except when it is convenient to mix their imports, with no regard for mutual compatibility. The conclusion is not that the diagrams are wrong; it is that they are maps, and the discipline of using a map is knowing where it ends. Every claim here is either an externally verified fact about the literature, or an explicitly marked argument; the premises the argument rests on — that a formalism can be about a physical reality at all, and that unchecked import-mixing is epistemically risky — are stated plainly in the final section, where the paper’s own map ends.

1. Introduction: the satisfaction of diagrams

There is a particular pleasure in watching a diagram close. A spreadsheet, a graph, a wiring diagram: connections assemble, the structure clicks, and something that was formless becomes shape. That satisfaction is real, and it is doing real work — in

engineering, in teaching, in communication. But the satisfaction is not evidence. A picture that feels right may still be a picture of nothing in particular, and the tighter its internal logic, the easier it is to forget that it is a picture at all.

This is the situation of the ZX calculus, the leading diagrammatic language of quantum computing. ZX diagrams are the workhorse of modern circuit optimization, error correction, and measurement-based computation. They are sound and complete for substantial fragments of quantum mechanics — results proved rigorously, not intuited [1, 2, 3, 4]. Their success is beyond dispute, and their utility is demonstrated daily in real compilers. Yet the same diagrams that are so carefully proved to capture the algebra of quantum computation are increasingly shown alongside objects that no proof in the calculus touches: spin networks from loop quantum gravity [5], quantum groups [6], holographic codes with Rényi entropy and toy black holes [7], lattice-surgery geometries [8]. Each of these imports arrives with its own rigorous story. What is never supplied is a check of whether the stories are compatible with each other — or with the 2-dimensional drawing plane on which they are all being told.

This paper is a fault-line audit. It takes the three named constructs of the ZX calculus — spiders, Pauli webs, and gadgets — and asks, for each: where did you come from, what do you silently carry, and where does the map end? It then audits the pairwise compatibility of the imports, and generalizes the resulting pattern into a “cafeteria problem” that is not specific to quantum computing at all: siloed disciplines remain separate except when it is convenient to mix and match, and the mixing happens without regard for mutual compatibility or contradiction.

The paper is deliberately narrow in its claims. It does not dispute the mathematics of the ZX calculus, which it imports and respects. It does not claim the diagrams are useless, or that their use is necessarily misleading — within the computational domain they are exactly as good as their theorems say. It claims one thing, precisely: that the physical bearing of the diagrams — their status as pictures of quantum reality rather than pictures of computation — is unestablished, that the published record shows imports from incompatible silos being loaded onto the diagrams without compatibility checks, and that this pattern is a general failure mode of cross-disciplinary borrowing. The evidence for the first two parts is in the literature and is cited. The third part is an argument, and it is marked as one.

2. The calculus and its constructs

The ZX calculus was introduced by Coecke and Duncan as a diagrammatic language for “interacting quantum observables” [9]. Its grammar is deceptively simple: two kinds of nodes — green spiders and red spiders — connected by wires, with angles attached to the nodes. A green spider represents the copying and merging of computational-basis information; a red spider the same operation in the Hadamard-rotated basis. The rewrite rules of the calculus — fusion, bialgebra, and their kin — let a practitioner transform one diagram into another while preserving the represented linear map exactly. Completeness, proved in stages by Backens [2], Jeandel, Perdrix, and Vilmart [1], and Wang [3], means that every matrix equality in the relevant fragment is provable diagrammatically: nothing is lost by working in the picture. Van de Wetering’s survey [10] is the standard working reference.

Three constructs do most of the work in modern applications.

Spiders. The generators themselves. A spider is a node with any number of inputs and outputs, carrying a phase. Its algebraic meaning is fixed: it is a special dagger Frobenius algebra — a structure that says “copy, then merge, coherently.” Spiders are what make the calculus a calculus: the fusion rule lets adjacent spiders combine, and the complementarity (bialgebra) rule captures the Z/X interplay that is the heart of quantum mechanics’ two non-commuting bases.

Pauli webs. When a ZX diagram is used to analyze a quantum error-correcting code, the stabilizer structure of the code can be tracked through the diagram as a “web” of Pauli operators — families of X/Z strings that commute with the code’s logical operations and whose behavior under the diagram’s rewrites reveals the code’s error properties. De Beaudrap and Horsman proved that the operations of the ZX calculus match the operations of surface-code lattice surgery exactly: red and green spiders correspond to rough and smooth merges and splits [8]. Pauli webs have since become the standard tool for reading stabilizer structure off diagrams, most strikingly in the holographic setting [7]. The calculus has also been extended to the infinite, translation-invariant codes that dominate modern error-correction proposals — lattice codes, convolutional codes — through the delayed stabilizer ZX-calculus, which adds a single delay generator feeding data between time steps [17].

Gadgets. Non-Clifford operations are the expensive part of quantum computation, and the diagrammatic community has developed “gadgets”: small diagrammatic patterns that encode a non-Clifford gate — the Hadamard gadget, the phase gadget — so that circuits can be optimized by rewriting around them. The machinery descends from the one-way quantum computer of Raussendorf, Browne, and Briegel [11], where computation proceeds by measuring entangled resource states, and it reaches its diagrammatic form in the ZH calculus of Backens and Kissinger [12], whose arity-generalized Hadamard nodes admit compact encodings of non-linear classical functions. Vandaele’s qubit-count optimization work [13] is a recent example of gadgetization in action: reversing the gadgetization of Hadamard gates to save qubits, with NP-hardness results and practical algorithms.

3. Import provenance: the cafeteria table

Every construct imports more than its definition. The following table gives, for each construct, its home discipline, its canonical sources, and — critically — what it silently carries that its users may not notice.

Construct	Home silo	What it imports	What it silently carries
Spiders	Categorical quantum mechanics (2D diagrammatic algebra)	Dagger-special Frobenius algebra structure; fusion; complementarity [9, 1, 2, 3]	The 2D topology of wires and planes; the completeness theorems prove facts about the <i>algebra</i> , nothing about spacetime
Pauli webs	Stabilizer formalism / quantum error	Stabilizer group structure; code distance and	A 2+1D code-over-time picture; in the holographic

Construct	Home silo	What it imports	What it silently carries
Gadgets	correction (lattice surgery, surface codes)	correction semantics; rough/smooth merge operations [8, 7]	application, an AdS/CFT vocabulary of entropy, wormholes, and black holes [7]
	Measurement-based computation / circuit optimization (compilation layer)	Graph-state resources; measurement-pattern semantics; T-count machinery [11, 12, 13]	The MBQC equivalence claims — a gadget is a map of a computation, validated by rewrite soundness, not by any physical realization

The pattern is already visible in this table. The three constructs come from three different disciplines — categorical algebra, error correction, and compilation — and in the computational domain they cooperate flawlessly, because they are all maps of the same algebraic object: a linear map between qubits. The cooperation is proved where it matters (soundness and completeness), and the compiler industry runs on it.

But the table also shows where the imports have started to outgrow the map. Spin networks from loop quantum gravity have been embedded into the ZXH calculus [5]. Braided ZX calculi have been built on the quantum group $u_q(\mathfrak{sl}_2)$ [6]. Holographic codes have been analyzed through Pauli webs, with Rényi entropy and toy black-hole/wormhole models computed diagrammatically [7]. These are imports from 3+1-dimensional particle physics and 1-dimensional thermodynamics — statistical mechanics, entropy, gravity — arriving on a 2-dimensional drawing plane.

4. Compatibility audit: where the map is coherent, and where it is not

The audit asks, for each pair of imports: are these two structures known to be mutually compatible, in the sense that the semantics they assign to the same diagram agree?

Within the computational domain, the answer is yes, with proof. Spiders and Pauli webs are coherent because the stabilizer fragment of the ZX calculus is complete [2, 14, 15]: every stabilizer equality is provable diagrammatically, the rule set has been shown minimal in the sense that no rewrite rule is redundant [18], and the lattice-surgery correspondence is an exact match, not an analogy [8]. Spiders and gadgets are coherent because gadgetization preserves circuit semantics — the ZH calculus is complete for universal quantum computation [12], and qubit-count optimization via gadget reversal preserves the number of non-Clifford gates [13]. Pauli webs and gadgets are coherent because both are compile-layer tools on the same stabilizer structure: the Pauli Fusion computational model, which can represent

lattice surgery operations, is natively depictable in ZX [13]. Within this domain, the diagrams are doing exactly what their theorems say they are doing. This is not in dispute, and the claim of this paper does not touch it.

At the boundary of the physical imports, the answer is: unchecked. Three specific fault lines emerge from the literature:

1. **The 2D-to-3+1D fault line.** Spin networks are structures of 3+1-dimensional quantum gravity — SU(2) representation theory carrying angular-momentum data [5]. Quantum groups carry deformation parameters that encode physical symmetry deformations [6]. When these are embedded in a 2-dimensional diagrammatic calculus, no published work checks whether the dimensional semantics of the source theory survive the embedding. The completeness theorems of the host calculus are theorems about its own algebra; they say nothing about the imported structures' physical claims.
2. **The 2D-to-1D fault line.** Holographic codes compute Rényi entropies and model black holes and wormholes through Pauli webs [7]. Entropy is a 1-dimensional thermodynamic concept — a number attached to a statistical ensemble — and the holographic dictionary that connects it to geometry is itself one of the most contested constructions in physics. The ZX diagrams import the vocabulary (entropy, wormholes, the AdS/CFT correspondence) without any demonstrated compatibility with the diagrammatic semantics. The same fault line appears one step earlier, in the diagrammatic treatment of probability itself: the decohered ZX-calculus extends the language to classical probability distributions over classical bits [16], and that classical-statistical fragment is routinely mixed with the quantum one without a boundary being drawn. The diagrams are not wrong to track stabilizers; they are unvetted as carriers of thermodynamic and statistical claims.
3. **The 3+1D-to-1D fault line inside the same diagram.** When spin networks [5] and holographic entropy [7] appear in the same diagrammatic tradition, the two physical imports are being combined without any argument that their respective physics is mutually consistent — the exact scenario the originating question named: imports from 3+1D particle physics and 1D thermodynamics mixed in a 2D picture.

A search of the literature for a published compatibility audit — a paper checking, across the silos, whether the imported semantics agree — found none as of August 2026. This absence is itself the finding: the fault lines are not disputed territory; they are unvisited territory.

5. The seam: where the map ends

A map is not wrong when it omits things; it is wrong when its users forget it omits things. Every construct has a seam — the place where the map stops being a reliable picture — and the seam is unmarked on the diagrams themselves.

- **The spider's seam.** A spider is a copy/merge operation on a measurement basis. It is not a particle; its wires are not worldlines; the drawing plane is not spacetime. The completeness theorems guarantee that every matrix equality is

provable in the picture — they guarantee nothing about whether the topology of the picture corresponds to the topology of any physical process. When a reader sees a spider as a physical event, they have crossed the seam.

- **The Pauli web’s seam.** A Pauli web is bookkeeping for stabilizer structure. Its QEC semantics — code distance, correction — are map-internal and sound. When a web is read as “the spacetime structure of the code,” or as evidence about black holes, the holographic vocabulary has been imported across the seam without a visa check.
- **The gadget’s seam.** A gadget is a diagrammatic encoding of a gate. Its MBQC equivalence claims are map-internal and proved. When a gadget is read as a hardware component — “this is how the T gate is built” — the physicalization assumptions of MBQC have been imported without the resource states being demonstrated.

Why does this matter? Because the cone of ignorance of a map is unknown by construction: a simplified map does not probe outside its own locality. The ZX calculus cannot tell you whether the 2D topology of its diagrams is the topology of quantum reality, because every question it can answer is answered inside the 2D picture. The seam is exactly where the map’s questions stop — and no theorem of the map can see past it.

6. The cafeteria problem

The pattern in the ZX literature is not local. Siloed disciplines — categorical algebra, stabilizer theory, measurement-based computation, particle physics, thermodynamics — remain separate in their journals, their vocabulary, and their standards of evidence. They come together only when it is convenient: when a diagram needs an entropy, a graviton, or a resource count, the import is fetched from the nearest silo and used without checking whether the silos’ claims are mutually compatible. The ZX calculus is a worked example of this cafeteria problem, but the mechanism is general:

1. **A map gains users because it is internally valid.** The better the map, the louder the silence about its edges.
2. **The seam is unmarked on the map.** Nothing on a ZX diagram tells you where the computational guarantee ends and the physical claim begins.
3. **Imports arrive unvetted.** The published record shows the mixing; no published record shows the compatibility check.
4. **The cone of ignorance is unknown.** Because the map never probes outside its own locality, nobody can say how much the map is missing — which is precisely the situation in which confident misuse thrives.

The cafeteria problem is a failure mode of *borrowing*, not of *building*. It is what happens when a successful notation travels faster than its caveats.

7. Map-aware practice

The remedy is not to abandon the diagrams. It is to use them as maps, with the seam declared:

- **Name the import provenance.** For every construct, know the home silo and what the import silently carries. The table in Section 3 is the template.
- **Mark the seam.** In any use of a ZX diagram that carries a physical claim — a particle, an entropy, a black hole — state where the diagram’s guarantees end.
- **Audit compatibility at the boundary.** When imports from different silos meet in one diagram, ask whether any published result establishes their mutual consistency. If not, say so.
- **Probe beyond the locality.** The map’s cone of ignorance is not probed by the map. It is probed by experiments and by formal arguments that connect the diagrammatic semantics to physical processes — the kinds of arguments that, as of this writing, the literature has not yet supplied for the physical imports.

None of this is anti-diagram. It is pro-honesty about what the diagram is.

8. Conclusion: where this paper’s map ends

The ZX calculus is one of the great successes of modern quantum computing — a rigorous, elegant, and genuinely useful picture of computation. Its spiders, Pauli webs, and gadgets are maps, and excellent ones, within their domain. This paper has argued that the physical bearing of those maps is unestablished, that the literature shows imports from 3+1D particle physics and 1D thermodynamics being loaded onto the 2D diagrams without compatibility checks, and that this cafeteria pattern is a general risk of cross-disciplinary borrowing. The argument is evidence- based where it cites the literature, and plainly argumentative where it generalizes.

It remains to say where the premises end — where this paper’s own map stops. Three things are assumed rather than derived, and the reader should weigh them accordingly. First, the representational stance: that a formalism can meaningfully be “about” a physical reality at all, so that questions of physical bearing are well-formed. Second, the epistemic-risk premise: that unchecked mixing of imports across silos is a real hazard, worth auditing, rather than a harmless byproduct of interdisciplinary work. Third, the map-territory distinction itself: that there is a difference between a picture of computation and a picture of reality, and that the difference matters. These are the unanalyzable primitives of this paper’s frame — the ground it stands on without proving. Everything else in the paper is either a verified fact about the literature or an argument built on that ground, and the boundary between the two has been marked throughout.

The satisfaction of a closing diagram is real. So is the discipline of remembering that it closed a map, not the world.

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